$$= \frac{144}{3} = \boxed{48} \text{ Ans}$$

Lec # 16 :-

ws : 13

$\Rightarrow$ Double integral of f can be written as the product of two single integrals.

$$\boxed{\iint_R g(x)\, h(y)\, dA = \int_a^b g(x)\, dx \times \int_c^d h(y)\, dy}$$

Example :- find double integral of $f(x,y) = \sin x \cos y$, if $R = \left[0, \frac{\pi}{2}\right] \times \left[0, \frac{\pi}{2}\right]$

Solution :-

$$= \int_0^{\pi/2} \int_0^{\pi/2} \sin x \cos y \, dx\, dy$$

$$= \int_0^{\pi/2} \sin x \, dx \times \int_0^{\pi/2} \cos y \, dy$$

$$= \left(-\cos x\right)\Big|_0^{\pi/2} \times \left(\sin y\right)\Big|_0^{\pi/2}$$

$$= \left[-\cos\left(\frac{\pi}{2}\right) - \left(-\cos(0)\right)\right] \times \left[\sin\left(\frac{\pi}{2}\right) - \sin(0)\right]$$

$$= \left[-0 + 1\right] \times \left[1 - 0\right] = 1 \times 1 = \boxed{1} \text{ Ans}$$

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Example:- $\int_1^5 \int_0^x (8x - 2y)\, dy\, dx.$

Solution:-

$= \int_1^5 \left[\int_0^x (8x - 2y)\, dy \right] dx$

$= \int_1^5 \left[8xy - \frac{\cancel{2}y^2}{\cancel{2}} \right]_0^x dx$

$= \int_1^5 \left[8xy - y^2 \right]_0^x dx$

$= \int_1^5 \left[\left(8x(x) - x^2 \right) - (0) \right] dx$

$= \int_1^5 (8x^2 - x^2)\, dx$

$= \int_1^5 7x^2\, dx$

$= \frac{7x^3}{3} \Big|_1^5$

$= \frac{7}{3} \left[5^3 - 1^3 \right]$

$= \frac{7}{3} \left[125 - 1 \right]$

$= \frac{7}{3} \left[124 \right] = \boxed{\frac{868}{3}}$ Ans.

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Ex:- $\int_0^2 \int_0^{y^2} x^2 y \, dx \, dy$

Solution:-

$$= \int_0^2 \left[\int_0^{y^2} (x^2 y) \, dx \right] dy$$

$$= \int_0^2 \left[\frac{x^3 y}{3} \right]_0^{y^2} dy$$

$$= \int_0^2 \left[\left(\frac{(y^2)^3 y}{3} \right) - (0) \right] dy$$

$$= \int_0^2 \left[\frac{y^7}{3} \right]$$

$$= \frac{y^7}{3} \Big|_0^2$$

$$= \frac{y^{7+1}}{3} \div 8$$

$$= \frac{y^8}{\cancel{8}3} \quad \frac{1}{8} \left[\frac{(2)^8}{3} \times \frac{1}{8} \right.$$

$$= \frac{\cancel{256}^{32}}{\cancel{24}_3} = \boxed{\frac{32}{3}} \quad \text{Ans.}$$

Date: 4-3-25 **Triple Integral** Day:

Over Rectangular Region

Evaluate triple integral.

Example:- $\iiint_B xyz \, dV$, where B is

Rectangular Box given by

$B = \{(x, y, z) \mid 0 \leq x \leq 1, \ -1 \leq y \leq 2, \ 0 \leq z \leq 3.$

Solution:-

$$= \int_0^3 \int_{-1}^2 \int_0^1 xyz \, dx \, dy \, dz$$

$$= \int_0^1 xyz \, dx = \frac{x^2}{2} yz \Big|_0^1$$

$$= \frac{1}{2} yz - 0$$

$$= \int_{-1}^2 \frac{1}{2} yz \, dy = \frac{1}{2} \frac{y^2}{2} z \Big|_{-1}^2$$

$$= \int \frac{1}{4} y^2 z \Big|_{-1}^2$$

$$= \left(\frac{1}{4} (2)^2 z\right) - \left(\frac{1}{4} (-1)^2 z\right)$$

$$= \left(\frac{1}{4} \times 4 \times z\right) - \left(\frac{1}{4} \times z\right)$$

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$$= \left[z - \frac{1}{4} z \right]$$

Taking L.C.M

$$= \frac{4z}{4} - \frac{z}{4} = \frac{4z - z}{4}$$

$$= \frac{3z}{4}$$

$$= \int_0^3 \frac{3z}{4} \, dz$$

$$= \left. \frac{3z^2}{4 \times 2} \right|_0^3$$

$$= \frac{3(3)^2}{8} - 0$$

$$= \frac{27}{8}$$ Ans.

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$$\int_0^{z}\int_0^{z^2}\int_0^{y-z} (2x-y)\,dx\,dy\,dz.$$

Solution:-

$$\int_0^{y-z} (2x-y)\,dx = \frac{\cancel{2}x^2}{\cancel{2}} - xy \Big|_0^{y-z}$$

$$= x^2 - xy = \left[(y-z)^2 - (y-z)y\right] - \left[0-0\right]$$

$$= \underbrace{(y-z)^2}_{a^2+2ab+b^2} - (y^2 - yz)$$

$$= \cancel{y^2} - 2yz + z^2 - \cancel{y^2} + yz$$

$$= -yz + z^2$$

$$= \int_0^{z^2} -yz + z^2\,dy$$

$$= -\frac{y^2}{2}z + yz^2 \Big|_0^{z^2}$$

$$= \left(-\frac{(z^2)^2 z}{2} + (z^2)z^2\right) - (0)$$

$$= \left(-\frac{z^4 z}{2} + z^4\right)$$

$$= -\frac{z^5}{2} + z^4.$$

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$$= \int_0^2 \left(\frac{-z^5}{2} + z^4 \right) dz$$

$$= \int_0^2 \left(\frac{-z^6}{6 \times 2} + \frac{z^5}{5} \right)$$

$$= \frac{-z^6}{12} + \frac{z^5}{5} \Big|_0^2$$

$$= \left(\frac{-(2)^6}{12} + \frac{(2)^5}{5} \right) - (0)$$

$$= \frac{-\cancel{64}^{32}}{\cancel{12}_6} + \frac{32}{5}$$

$$= \frac{-\cancel{32}^{16}}{\cancel{6}_3} + \frac{32}{5}$$

$$= \frac{-16}{3} + \frac{32}{5}$$

$$= \frac{-16 \times 5}{15} + \frac{32 \times 3}{15}$$

$$= \frac{-80 + 96}{15} = \boxed{\frac{16}{15}}$$ Ans.

Date: Day: M T W T F S

(i) $\iiint_E (xy + z^2)\, dV$

$E = \{(x, y, z) \mid 0 \leq x \leq 2,\ 0 \leq y \leq 1,\ 0 \leq z \leq 3$

Solution:-

$= \int_0^3 \int_0^1 \int_0^2 (xy + z^2)\, dx\, dy\, dz$

$= \int_0^2 \left(\frac{x^2 y}{2} + xz^2\right) dx$

$= \frac{x^2 y}{2} + xz^2 \Big|_0^2 \quad = \left(\frac{(2)^2}{2} y + (2)z^2\right) - (0)$

$= \frac{\cancel{4}^{2}}{\cancel{2}} y + 2z^2$

$= 2y + 2z^2$

$= \int_0^1 2y + 2z^2\, dy$

$= \frac{2y^2}{2} + 2yz^2 \Big|_0^1 \quad = \left(\frac{2(1)^2}{2} + 2(1)z^2\right) - (0)$

$= \frac{\cancel{2}}{\cancel{2}} + 2z^2 \quad = 1 + 2z^2$

Date:

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$$= \int_0^3 1 + 2z^2 \, dz$$

$$= \int_0^3 \left(z + \frac{2z^3}{3} \right)$$

$$= z + \frac{2z^3}{3} \Big|_0^3$$

$$= \left((3) + \frac{2(3)^3}{3} \right) - (0)$$

$$= (3) + \frac{2(27)}{3}$$

$$= 3 + \frac{54}{3}$$

$$= \frac{9+54}{3} = \frac{\cancel{63}^{21}}{\cancel{3}}$$

$\boxed{= 21}$ Ans.